

The Higgs Domain Endpoint: Carrier, Identity, Scale, Vacuum, Extension, and Projection

Paper 6 of the Higgs Structural Endpoint Arc

Amos Jay Maley

June 2026

Abstract

This paper closes the six-paper Higgs structural endpoint arc. The preceding manuscripts establish the Higgs bridge/fixation carrier C_H , fix Higgs structural identity as admissible occupation of that carrier, delete primitive Higgs-scale standing, classify Higgs-vacuum claims by route availability and standing closure, and assign extension claims by carrier relation. The present paper is not a retrospective summary. It proves the domain endpoint that those closures jointly determine: no residual same-scope primitive Higgs attribute remains outside the carrier architecture.

The paper defines a unified Higgs normal form

$$R_H^* = (D_H, C_H, H_{\text{role}}, R_H^{\text{id}}, R_H^{\text{scale}}, R_H^{\text{vac}}, R_H^{\text{ext}}, \Pi_{\text{met}}, \text{Cert}_H^*, \rho_H^*)$$

and a final status classifier for submitted Higgs-domain attribute claims. Every same-scope standing-bearing attribute claim factors through R_H^* . Scalarity, vev-status, potential-minimum status, pole mass, running scale, hierarchy comparison, lower branch, metastability, thermal history, electroweak-baryogenesis route, extra scalar content, composite realization, portal structure, and metric comparison are not discarded; they are typed. They are registered as licensed representatives, measurements, carrier-indexed scale statuses, vacuum-route statuses, extension dispositions, metric projections, target updates, counterexample-burden submissions, or failed primitive claims. Counterexample burden is not a survivor status.

The endpoint theorem is

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Attr}_H \setminus C_H) = 0.$$

Equivalently, the Higgs-domain primitive is $C_H = H_{\text{br}/\text{fix}}$, and all same-scope Higgs-domain standing claims are carrier-grounded or registered downstream of that carrier. The paper makes no numerical prediction of Higgs masses, widths, couplings, branching ratios, self-couplings, vacuum lifetimes, or BSM spectra. Its result is the closed status architecture for Higgs-domain attributes after the carrier, identity, scale, vacuum, and extension routes have been exhausted.

Keywords: Higgs; AASC; endpoint synthesis; bridge/fixation carrier; structural identity; no primitive attribute; scale standing; vacuum route status; extension classifier; metric projection; representative collapse.

Read this first: endpoint proof spine

The proof has six layers.

Layer I: inherited closures. Paper 1 supplies the carrier package $(D_H, C_H, H_{\text{role}}, R_H, \text{SF}_H)$. Paper 2 fixes $\text{ld}_H = \text{Occ}_{D_H}(C_H)$. Paper 3 deletes primitive scale standing. Paper 4 classifies vacuum and

thermal claims as route statuses. Paper 5 classifies extension claims by carrier relation. This paper does not reprove those modules.

Layer II: unified attribute target. A submitted Higgs-domain attribute claim $\text{Attr}_H(a)$ is any claim that purports to give standing to identity, scale, vacuum, representative, measurement, thermal, extension, metric, or unknown Higgs-associated content. It is evaluated in the fixed target D_H .

Layer III: unified normal form. Standing-bearing attribute use must descend to R_H^* . The normal form is not a new primitive; it is the closure product of R_H^{id} , R_H^{scale} , R_H^{vac} , R_H^{ext} , and the projection layer.

Layer IV: attribute-status exhaustion. The final classifier is layered. A submitted attribute has a raw descriptive mode, a module-entry status, a primary report-level disposition, and optional ancillary tags. Target updates and counterexample burdens are submitted-claim outcomes, not same-scope standing forms. Carrier, representative, measurement, scale status, vacuum route status, extension disposition, and metric projection are the possible same-scope report roles.

Layer V: residual primitive deletion. Any alleged residual submitted Higgs attribute is identity-like, scale-like, vacuum-like, extension-like, metric-like, representative-like, target-update, or untyped. The first four are classified by Papers 2-5, metric-like claims are projections, representative-like claims are licensed descriptions, target-update submissions leave same scope, and untyped claims fail. No residual primitive locus remains.

Layer VI: endpoint. The closed domain endpoint is $\text{Endpoint}_H(D_H) = (C_H, \text{Rep}_H, R_H^{\text{scale}}, R_H^{\text{vac}}, R_H^{\text{ext}}, \Pi_{\text{met}}, \text{Update}_H, \text{Fail}_H)$. The same-scope standing interior is carried by C_H and licensed representatives, measurements, scale statuses, vacuum-route statuses, extension dispositions, and projections; updates and failures are registered exits or defeats. This is the Higgs-domain structural endpoint, not a replacement for standard Higgs phenomenology.

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1 Introduction: from papers to endpoint architecture

The earlier papers in this arc close the major routes by which primitive Higgs standing could be claimed. The bridge paper establishes the standing-bearing carrier. The identity paper proves that Higgs identity is unique admissible occupation of that carrier. The scale paper deletes primitive Higgs-scale standing. The vacuum paper separates route availability from standing closure. The extension paper classifies extra scalar, composite, portal, and role-splitting claims by relation to the carrier. The present paper assembles those closures into a domain endpoint.

The point of a synthesis theorem is not to restate the earlier papers. Its job is to answer the remaining question: after identity, scale, vacuum, extension, and metric projection have been typed, can a residual primitive Higgs attribute still be asserted in the same target? The answer is no. A submitted Higgs-domain attribute claim either is the carrier C_H , is a licensed representative of C_H , is a measured or carrier-indexed scale claim, is a vacuum-route status, is an extension disposition, is a projection display, is a declared target update, is a counterexample-burden submission, or fails as a primitive claim.

The theorem is structural. It does not predict collider observables. It does not decide whether a future experiment will measure a new scalar, refine the Higgs width, constrain the self-coupling, detect a portal, infer a composite sector, or sharpen vacuum-stability inputs. It says how such attributes enter the standing architecture. A future measured attribute is not outside the endpoint; it is classified by the endpoint.

Nonclaim 1.1 (No phenomenological replacement). The paper does not replace Standard Model Higgs terminology, collider measurement, effective-field-theory analysis, vacuum-stability calculation, or BSM model building. It classifies the standing role of Higgs-domain claims relative to the carrier architecture.

Nonclaim 1.2 (No numerical prediction). The paper does not predict numerical Higgs masses, widths, self-couplings, branching ratios, Yukawa values, total width, vacuum lifetime, transition temperature, gravitational ratio, or BSM spectra.

2 Dependency map and imported endpoint package

The proof depends on the local outputs of Papers 1-5. It does not import the full AASC kernel afresh. The dependency lock is shown in Table 1.

Table 1: Imported closures for Paper 6.

Paper	Imported closure	Use in Paper 6
Paper 1	$D_H, C_H, H_{\text{role}}, R_H, \text{SF}_H$	Supplies the carrier, same-scope target, role bundle, representative-collapse discipline, and standing filter.
Paper 2	$\text{Id}_H = \text{Occ}_{D_H}(C_H)$ and R_H^{id}	Deletes primitive representative identity grounds and fixes identity through carrier occupation.

Paper	Imported closure	Use in Paper 6
Paper 3	R_H^{scale} and $\text{Surv}_{D_H}^{\text{prim}}(\text{Scale}_H) = 0$	Classifies vev, mass, running, hierarchy, cutoff, and metric-ratio claims as carrier-indexed scale statuses or projections.
Paper 4	R_H^{vac} , route availability, and standing-closure distinction	Classifies lower branch, metastability, instability crossing, lifetime, thermal, and EWBG claims as route statuses or target updates.
Paper 5	R_H^{ext} and extension classifier	Classifies extra scalars, composites, portals, supersymmetric sectors, pNGB presentations, and role splitting by carrier relation.
Fixed-domain support	no fifth governance-bearing locus	Used to close the residual primitive-attribute classifier.
Impossibility support	no independent standing generators, no cross-domain standing transfer, no same-act repairs	Blocks measurement success, route availability, extension usefulness, or projection display from generating primitive standing.

Definition 2.1 (Inherited Higgs endpoint package). The inherited package for Paper 6 is

$$\mathcal{P}_H^* = (D_H, C_H, H_{\text{role}}, R_H^{\text{id}}, R_H^{\text{scale}}, R_H^{\text{vac}}, R_H^{\text{ext}}, \text{SF}_H).$$

It is the fixed source package from which the unified normal form is constructed.

Lemma 2.2 (Package adequacy). *The inherited package $\mathcal{P}_H^*$, together with the cumulative exhaustion register E_H of Section 7, contains the module needed to classify every submitted Higgs-domain attribute function in the fixed target: carrier occupation, identity redescription, representation, measurement, scale, vacuum route, extension disposition, projection, target update, counterexample burden, and primitive failure.*

Proof. Paper 1 supplies the carrier and same-scope target. Paper 2 covers identity claims. Paper 3 covers scale and hierarchy claims. Paper 4 covers vacuum, lower-branch, thermal, and route-availability claims. Paper 5 covers extra-scalar and extension claims. Metric comparison is not a separate identity or scale generator; it is a projection layer introduced after carrier and scale status. The exhaustion register records the residual routes after these imports. Hence a submitted attribute function either enters one imported module, enters the projection layer, declares target update, is placed under counterexample burden, or fails as a primitive assertion. $\square$

3 Fixed target and same-scope discipline

Definition 3.1 (Higgs-domain attribute claim). A Higgs-domain attribute claim, written $\text{Attr}_H(a)$, is a submitted claim that some attribute a fixes, carries, explains, modifies, measures, represents, or projects Higgs-domain standing in D_H . Attribute claims include identity, scale, representative, measurement, vacuum, thermal, route, extension, metric, and unknown-attribute claims.

Definition 3.2 (Same-scope attribute claim). An attribute claim a is same-scope in D_H iff it preserves:

1. the target D_H ,
2. the carrier C_H ,
3. the role bundle H_{role} ,
4. identity redescription through R_H^{id} ,
5. scale-standing discipline through R_H^{scale} ,
6. vacuum-route discipline through R_H^{vac} ,
7. extension-status discipline through R_H^{ext} , and
8. the admissibility boundary and invariant bundle inherited from the local carrier package.

If any of these is altered as part of the attribute claim, the claim is a target update rather than a same-scope survivor.

Criterion 3.3 (Same-scope discipline). A claim does not remain same-scope by using Higgs vocabulary. It remains same-scope only by preserving the carrier package and the downstream normal forms. In particular, adding scalar inventory, changing the invariant bundle, replacing the carrier, altering the comparison target, or importing a new admissibility boundary is **TargetUpdate**, not same-scope refinement.

Proposition 3.4 (Target-update separation). *If $\text{Attr}_H(a)$ changes carrier inventory, invariant bundle, comparison target, or admissibility boundary, then a is not a same-scope residual primitive attribute. It is classified as **TargetUpdate**.*

Proof. The same-scope discipline fixes D_H , C_H , H_{role} , and the downstream normal forms. A claim that alters those items changes the object being evaluated. Such a claim may be physically meaningful in a richer target, but it is not a same-scope primitive survivor in D_H . $\square$

4 Unified Higgs normal form

Definition 4.1 (Unified Higgs normal form). The unified Higgs normal form is

$$R_H^* = (D_H, C_H, H_{\text{role}}, R_H^{\text{id}}, R_H^{\text{scale}}, R_H^{\text{vac}}, R_H^{\text{ext}}, \Pi_{\text{met}}, \text{Cert}_H^*, \rho_H^*).$$

Here Π_{met} is the metric/projection display layer, Cert_H^* is the standing certificate appropriate to the submitted attribute class, and ρ_H^* is the final status classifier.

Definition 4.2 (Same-scope and update normalizers). The same-scope partial normalizer $\text{NF}_{H,\text{same}}^*(a)$ is defined exactly when a submitted attribute claim a preserves D_H , C_H , H_{role} , and the inherited invariant bundle; enters the appropriate inherited normal form R_H^{id} , R_H^{scale} , R_H^{vac} , R_H^{ext} , or the projection layer Π_{met} ; and receives a same-scope report role under ρ_H^* . The target-update partial normalizer $\text{NF}_{H,\text{update}}^*(a)$ is defined exactly when a declares a changed carrier inventory, invariant bundle, comparison target, or admissibility boundary and is therefore evaluated as a richer-scope update rather than as a same-scope survivor.

Remark 4.3 (Why the normalizers are split). A target update may be a legitimate submitted outcome, but it is not a same-scope standing form. The notation NF_H^* is used only for the combined registry of the two partial normalizers; standing-bearing same-scope claims use $\text{NF}_{H,\text{same}}^*$, and richer-scope/update claims use $\text{NF}_{H,\text{update}}^*$.

Theorem 4.4 (Same-scope unified normal-form theorem). *For every same-scope standing-bearing Higgs-domain attribute claim,*

$$\text{Stand}_{D_H}(\text{Attr}_H(a)) = 1 \implies \text{NF}_{H,\text{same}}^*(a) \downarrow.$$

If a submitted claim is a target update, then $\text{NF}_{H,\text{update}}^(a) \downarrow$ and a is not a same-scope survivor.*

Proof. Let a be a same-scope standing-bearing Higgs-domain attribute claim. Same-scope standing means that a preserves D_H , C_H , H_{role} , and the inherited invariant bundle. If it is identity-like, the Paper 2 normal form R_H^{id} is required. If it is scale-like, the Paper 3 normal form R_H^{scale} is required. If it is vacuum-like, thermal, or route-like, the Paper 4 normal form R_H^{vac} is required. If it is extension-like while preserving C_H , the Paper 5 same-scope extension normal form R_H^{ext} is required. If it is metric-like, it is a projection layer Π_{met} rather than a primitive route. If it is representative-like, it must be licensed by C_H . In every same-scope case the claim enters $\text{NF}_{H,\text{same}}^*$. If the claim alters carrier inventory, invariant bundle, comparison target, or admissibility boundary, it does not preserve same scope and therefore enters $\text{NF}_{H,\text{update}}^*$ instead. $\square$

Corollary 4.5 (No direct standing outside the same-scope normal form). *If $\text{NF}_{H,\text{same}}^*(a)$ is undefined and a is not a target update with $\text{NF}_{H,\text{update}}^*(a) \downarrow$, then a has no same-scope standing-bearing Higgs-domain status.*

5 Attribute status taxonomy

Definition 5.1 (Layered final status classifier). For a submitted attribute claim a , the final classifier has three layers:

$$\rho_H^*(a) = (\rho_{H,\text{raw}}^*(a), \rho_{H,\text{module}}^*(a), \rho_{H,\text{disp}}^*(a); \text{Anc}_H^*(a)).$$

The raw layer records the descriptive mode of the claim; the module layer records which inherited normal form or update branch evaluates it; the disposition layer records the primary report-level status; and $\text{Anc}_H^*(a)$ records optional ancillary tags when ordinary Higgs language has several roles. The possible primary dispositions are exactly the status classes listed in Table 2. The classifier reports the standing role of the claim, not its empirical truth or phenomenological value.

Table 2: Final Higgs-domain status classes.

Status	Meaning	Typical entries
Carrier	The standing-bearing Higgs-domain primitive.	$C_H = H_{\text{br/fix}}$.
Representative	Licensed description of C_H , not an identity ground.	scalar field language, vev notation, potential chart, field-coordinate description.
Measurement	Empirical witness or measured representative.	pole mass, width constraint, coupling measurement, branching-ratio measurement.
ScaleStatus	Carrier-indexed scale standing or scale classifier.	vev scale, running mass, hierarchy comparison, cutoff sensitivity, instability crossing as scale marker.
VacuumRouteStatus	Vacuum, lifetime, thermal, or route-status classification.	lower branch, metastability, crossover, EWBG route, lifetime certificate.

Status	Meaning	Typical entries
ExtensionDisposition	Extension disposition relative to C_H .	additional non-Higgs field, carrier-preserving enrichment, role-split reassembly.
MetricProjection	Projection or display of carrier-indexed structure in metric or gravity language.	Planck comparison, metric-envelope display, faithful projection ratio.
TargetUpdate	Declared richer-scope or changed-domain claim.	richer-scope update, new scalar inventory changing carrier boundary, UV completion, altered invariant bundle.
CounterexampleBurden	Submitted same-scope non-reducing survivor burden, not a standing survivor.	alleged new primitive attribute outside all registered classes.
FailedPrimitive	Primitive claim with no standing locus.	bare scalarity as identity, isolated vev as primitive scale, lower branch as primitive endpoint.

Table 3: Layer distinction in the final classifier.

Layer	What it records	Why it is not a rival to the other layers
Raw descriptive mode	The surface content: measurement, scale, vacuum, extension, projection, representative, and so on.	One claim may have several raw descriptors in ordinary language.
Module status	The inherited normal form or update/burden review that evaluates the claim.	The module supplies the audit route; it is not the final standing disposition.
Final report-level disposition	The registered endpoint status used in the theorem.	Exactly one disposition is assigned after the boundary/update and module checks.

Theorem 5.2 (Layered classifier totality). *Every submitted Higgs-domain attribute claim receives one raw descriptive mode, one module-entry status, one primary report-level disposition, and a finite set of ancillary tags. The primary disposition is unique; ancillary tags record non-governing descriptive overlap.*

Proof. The decision procedure is deterministic. First, if the claim changes D_H , C_H , H_{role} , or the invariant bundle, the module entry is the update branch and the primary disposition is **TargetUpdate**. Second, if it is a submitted faithful non-reducing survivor challenge, the primary disposition is **CounterexampleBurden**, a burden state rather than a standing survivor. Third, if it preserves same scope, it is routed by governing standing function: carrier occupation, representative description, measurement witness, scale status, vacuum route status, extension disposition, or metric projection. A pole mass can therefore be both a measurement and scale-relevant content, but only one role is primary for the submitted use; the other is recorded in $\text{Anc}_H^*(a)$. If no route applies and the claim asserts primitive standing, the disposition is **FailedPrimitive**. Thus the layered classifier is total without forcing multi-role language into a false single-role ontology. $\square$

6 Attribute-function exhaustion

A residual primitive can be alleged only if it performs some Higgs-domain standing function not already assigned by the final classifier. The relevant functions are exhausted in Table 4. The table is a standing-function ledger, not a list of physics models.

Table 4: Attribute-function exhaustion.

Function of the claim	Fixed-domain locus	Registered result
Ground the Higgs carrier	carrier occupation	Carrier, uniquely C_H .
Fix what counts as the Higgs	identity redescription	R_H^{id} or failed primitive identity.
Describe, coordinate, or present the carrier	representative quotient	Representative.
Report empirical response	measurement witness	Measurement.
Fix scale, magnitude, running, or hierarchy	scale normal form	ScaleStatus or target update.
Classify vacuum, lower branch, lifetime, thermal route, or EWBG route	vacuum route normal form	VacuumRouteStatus, target update, or failure.
Add scalar or sector structure	extension carrier-relation classifier	ExtensionDisposition, TargetUpdate, or failure.
Display in metric or gravitational terms	projection layer	MetricProjection or target update.
Alter domain or invariant bundle	boundary typing	TargetUpdate.
Assert an unregistered primitive	no same-scope standing locus	CounterexampleBurden if faithful burden is submitted; otherwise FailedPrimitive.

Lemma 6.1 (No eleventh attribute function). *No same-scope Higgs-domain attribute function remains outside the function ledger of Table 4.*

Proof. A submitted Higgs-domain attribute claim can ground, identify, describe, measure, scale, route, extend, project, declare target update, or assert primitive standing. Among same-scope claims, the standing functions are the interior loci fixed by the inherited carrier package and downstream normal forms; target update exits the fixed target, and primitive assertion is routed to burden or failure. A purported additional same-scope function must change standing while changing none of carrier occupation, identity redescription, representative quotient, measurement witness, scale status, vacuum route, extension disposition, projection layer, or target boundary. Fixed-domain exhaustion excludes such a locus. If the function changes one of those items, it reduces to the listed rows; if it changes none, it is bookkeeping. $\square$

7 The exhaustion register E_H

Definition 7.1 (Cumulative exhaustion register). The cumulative Higgs exhaustion register is

$$E_H = (E_{\text{id}}, E_{\text{scale}}, E_{\text{vac}}, E_{\text{ext}}, E_{\text{proj}}, E_{\text{update}}, E_{\text{burden}}, E_{\text{fail}}).$$

It records eliminated primitive claim forms and the registered statuses that remain.

Table 5: Cumulative exhaustion register.

Register component	Primitive claim eliminated	Registered result
E_{id}	scalarity, vev-status, potential minimum, pole mass, RG label, thermal role, metric comparison as identity ground	$\text{Id}_H = \text{Occ}_{D_H}(C_H)$; representatives licensed by C_H .
E_{scale}	isolated vev scale, bare mass, cutoff scale, running crossing, raw Planck ratio as primitive scale	carrier-indexed scale status and conditional projection display.
E_{vac}	lower branch, metastability, lifetime route, thermal transition, EWBG route as primitive Higgs ground	route availability, certificate-closed carrier-indexed vacuum content, or failure; changed-domain submissions go to E_{update} .
E_{ext}	extra scalar, second doublet, compositeness, portal, pNGB, SUSY, role-split sector as automatic extra Higgs identity	extension disposition by carrier relation; changed-domain submissions go to E_{update} .
E_{proj}	metricity, gravity comparison, hierarchy display as primitive Higgs foundation	downstream projection layer Π_{met} .
E_{update}	carrier inventory, invariant bundle, comparison target, or admissibility boundary alteration submitted as same-scope survivor	target-update normal form, not same-scope standing.
E_{burden}	non-reducing residual survivor asserted after all registered routes are exhausted	counterexample burden as review state, not survivor status.
E_{fail}	residual untyped primitive attribute outside all registered routes	FailedPrimitive.

Lemma 7.2 (Register coverage). *Every submitted attribute class in the arc is covered by one component of E_H .*

Proof. Identity-like attributes are covered by E_{id} ; scale-like attributes by E_{scale} ; vacuum, thermal, and route attributes by E_{vac} ; extension attributes by E_{ext} ; metric or hierarchy-display attributes by E_{proj} .

If an alleged attribute changes the target, it is covered by E_{update} . If it asserts a faithful non-reducing same-scope survivor after all registered routes are exhausted, it is covered by E_{burden} . If it is untyped within D_H , it is covered by E_{fail} . No submitted class remains outside the register. $\square$

8 No residual primitive attribute theorem

Definition 8.1 (Residual primitive Higgs attribute). A residual primitive Higgs attribute is a submitted attribute claim a such that $\text{Attr}_H(a)$, $a \neq C_H$, and a purports to carry primitive same-scope Higgs standing while not entering R_H^{id} , R_H^{scale} , R_H^{vac} , R_H^{ext} , or Π_{met} , and while not declaring a target update or accepting failed-primitive classification.

Lemma 8.2 (Residual primitive classification). *A residual primitive Higgs attribute cannot remain unclassified while preserving D_H .*

Proof. Let a be a submitted residual primitive attribute. If it fixes what the Higgs is, it is identity-like and enters R_H^{id} . If it fixes a scale, magnitude, hierarchy, or running claim, it enters R_H^{scale} . If it fixes a vacuum, lower branch, lifetime, thermal, or route claim, it enters R_H^{vac} . If it concerns additional scalar, sector, compositeness, role splitting, or BSM update, it enters R_H^{ext} or TargetUpdate . If it compares to metric or gravity structure, it enters Π_{met} . If it refuses every registered route while claiming same-scope primitive standing, it is a counterexample-burden submission or a failed primitive assertion. No unclassified same-scope locus remains. $\square$

Theorem 8.3 (No residual primitive Higgs attribute). *In the fixed Higgs-domain target,*

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Attr}_H \setminus C_H) = 0.$$

Proof. Assume for contradiction that some $a \neq C_H$ survives as a same-scope primitive Higgs attribute. By the residual primitive classification lemma, the submitted claim is identity-like, scale-like, vacuum-like, extension-like, metric-like, a target-update submission, a counterexample-burden submission, or failed primitive. Identity-like primitive claims outside C_H are deleted by Paper 2. Scale-like primitive claims are deleted by Paper 3. Vacuum-like primitive claims are deleted or routed by Paper 4. Extension-like primitive claims are classified by Paper 5 and do not stand outside C_H without a discharged faithful counterexample. Metric-like claims are projections, not primitive grounds. Target-update submissions are not same-scope survivors. Counterexample-burden submissions are not survivors. If such a burden were discharged, it would break or revise an imported module, or define a new target, rather than survive inside the already closed endpoint package. Failed primitive claims do not survive. Therefore, relative to the imported package, no such a survives. $\square$

Corollary 8.4 (Carrier uniqueness at the domain endpoint). *The only same-scope Higgs-domain primitive surviving the six-paper arc is C_H .*

9 Higgs domain endpoint theorem

Theorem 9.1 (Higgs domain endpoint). *The closed Higgs-domain endpoint is*

$$\text{Endpoint}_H(D_H) = (C_H, \text{Rep}_H, R_H^{\text{scale}}, R_H^{\text{vac}}, R_H^{\text{ext}}, \Pi_{\text{met}}, \text{Update}_H, \text{Fail}_H).$$

Every submitted Higgs-domain attribute claim is classified as same-scope, target-update, counterexample-burden, or primitive-failed. Every same-scope standing-bearing Higgs-domain claim is the carrier C_H ,

a licensed representative or measurement of C_H , a carrier-indexed scale status, a vacuum route status, an extension disposition, or a metric projection. Target updates and primitive failures are recorded endpoint outcomes, not same-scope standing forms. Counterexample burden is tracked by the classifier and exhaustion register as a review state, not as an endpoint component. No unregistered primitive Higgs attribute remains.

Proof. The unified same-scope normal-form theorem shows that every standing-bearing same-scope attribute claim enters R_H^* through $\text{NF}_{H,\text{same}}^*$. The target-update normal-form separation theorem records claims that leave the fixed target through $\text{NF}_{H,\text{update}}^*$. The layered classifier totality theorem assigns a registered final disposition to every submitted claim. The cumulative exhaustion register covers identity, scale, vacuum, extension, projection, update, burden, and failure routes. The no residual primitive Higgs attribute theorem deletes the only remaining alternative: a primitive same-scope attribute outside the carrier architecture. Thus the displayed endpoint tuple exhausts the fixed Higgs-domain target while keeping same-scope standing distinct from update and failure outcomes. $\square$

Corollary 9.2 (No attribute laundering). *A Higgs-associated measurement, representative, route, extension success, metric display, or model-building utility cannot launder itself into primitive same-scope Higgs standing. It must retain the status assigned by ρ_H^* .*

10 Unknown attributes

The endpoint theorem does not make future measurement unnecessary. It makes future attributes classifiable. A future attribute may be a new measurement of a licensed representative, a carrier-preserving enrichment, a scale-status refinement, a vacuum-route certificate, an extension disposition, a metric projection, a target update, or a counterexample-burden submission. It is not automatically a residual primitive.

Table 6: Unknown-attribute classifier.

Unknown attribute form	Standing question	Endpoint status
New decay or branching-ratio information	Does it measure a representative of C_H ?	Measurement.
Higgs self-coupling refinement	Does it refine carrier-indexed dynamics without grounding identity?	Measurement or ScaleStatus, depending on use.
New scalar resonance	Does it preserve C_H , add a non-Higgs field, or change the target?	ExtensionDisposition or TargetUpdate.
Composite evidence	Does the composite presentation preserve the carrier role?	ExtensionDisposition; possibly TargetUpdate.
Vacuum-stability refinement	Does it close a lifetime/stability certificate?	VacuumRouteStatus.
Thermal-transition evidence	Does it close a thermal route in the same target or in a richer target?	VacuumRouteStatus or TargetUpdate.

Unknown attribute form	Standing question	Endpoint status
Hierarchy or gravity comparison	Is it a faithful projection of carrier-indexed standing?	MetricProjection or ScaleStatus.
Claimed new primitive Higgs attribute	Does it preserve D_H and evade every classifier without reducing to C_H ?	CounterexampleBurden as burden state; otherwise FailedPrimitive.

Theorem 10.1 (Unknown attributes are classifier inputs). *A future Higgs-associated attribute is not outside the endpoint architecture. It enters ρ_H^* unless it declares a new target outside D_H .*

Proof. The classifier is defined on submitted Higgs-domain attribute claims, not only on known attributes. A future claim still has a standing role: representative, measurement, scale, vacuum route, extension, projection, target update, counterexample burden, or primitive failure. If it declares a new target, it is a target update. Otherwise it is classified in D_H . $\square$

11 Metricity as projection surface

Metric comparison has a legitimate role, but it is not foundational to the Higgs endpoint. The metric or gravitational display layer appears after carrier, identity, scale, and comparison discipline have been fixed. In the notation of the arc,

$$\Pi_{\text{met}}(C_H)$$

is a projection or display of carrier-indexed structure. It is not an independent Higgs primitive and not the source of Higgs identity.

Definition 11.1 (Metric projection). A metric projection is a map, display, or comparison surface that reports carrier-indexed Higgs structure in metric, gravitational, hierarchy, Planck-scale, or closure-scale language. It is faithful only when it preserves the carrier relation and does not import an empirical or representative anchor as primitive standing.

Theorem 11.2 (Metric projection non-primitivity). *No metric or gravity comparison supplies primitive same-scope Higgs standing. Every admissible such claim is a projection, carrier-indexed scale status, or target update.*

Proof. If the comparison preserves C_H and the inherited scale-standing discipline, it enters Π_{met} or R_H^{scale} . If it alters the comparison target, gravitational carrier, or invariant bundle, it is a target update. If it treats metricity, Planck labels, closure-scale display, or hierarchy ratio as primitive Higgs identity or scale ground, it violates the inherited carrier and scale results and fails as primitive standing. $\square$

12 Counterexample burden

Definition 12.1 (Faithful residual counterexample). A faithful residual counterexample to the domain endpoint is a submitted attribute claim x satisfying all of the following:

1. x preserves D_H ;
2. x preserves H_{role} ;

3. x does not reduce to C_H ;
4. x does not rely on scalarity, vev, mass, potential, running scale, vacuum branch, thermal route, extension success, or metric projection as primitive ground;
5. x does not declare `TargetUpdate`;
6. x survives the Paper 2 identity classifier, Paper 3 scale classifier, Paper 4 vacuum-route classifier, Paper 5 extension classifier, and the projection classifier of this paper.

Theorem 12.2 (Counterexample burden theorem). *Any submitted residual primitive attribute that fails one condition of the faithful counterexample definition is classified as Representative, Measurement, ScaleStatus, VacuumRouteStatus, ExtensionDisposition, MetricProjection, TargetUpdate, or FailedPrimitive. A submission satisfying the burden conditions is classified as CounterexampleBurden. If the preservation and non-reduction proof is supplied, the result is not an additional survivor inside the closed endpoint; it is a challenge to an imported module or a declaration of a new target requiring a revised endpoint package.*

Proof. Failure of carrier preservation, target preservation, or invariant-bundle preservation produces target update or failure. Reliance on representative, scale, vacuum, thermal, extension, or metric features routes the claim through the corresponding classifier. Failure to survive an inherited classifier produces that classifier’s outcome. A submission that satisfies the stated conditions has not yet produced a standing survivor; it has only identified the burden that a same-scope counterexample must discharge. Therefore `CounterexampleBurden` is a burden state, not a survivor status, and a discharged burden revises the premises of the closed package rather than living inside it as a hidden eleventh status. $\square$

13 Relation to standard Higgs language

The endpoint theorem preserves ordinary Higgs language by typing it. It does not remove the Higgs field from QFT descriptions, deny the utility of the vev, reject the Higgs pole mass, erase effective potentials, dismiss RG running, or forbid extended scalar sectors. It prevents these descriptions from functioning as primitive standing grounds after the carrier architecture has been fixed.

Table 7: Standard Higgs language under the endpoint classifier.

Standard phrase	Preserved role	Blocked promotion
“the Higgs field”	representative field description of the carrier	primitive identity by scalarity alone
“the Higgs vev”	licensed scale or vacuum representative	primitive identity or primitive scale ground
“the Higgs mass”	measured excitation/scale content	primitive carrier or identity ground
“the Higgs potential”	route and representative chart	primitive vacuum endpoint or identity ground
“Higgs vacuum stability”	vacuum-route status requiring certificates	primitive lower-branch standing

Standard phrase	Preserved role	Blocked promotion
“extended Higgs sector”	extension claim requiring carrier-relation classification	automatic same-scope extra Higgs identity
“Higgs hierarchy”	carrier-indexed scale comparison or projection	primitive scale mismatch independent of C_H
“Higgs and gravity”	metric projection or target update	metricity as foundation of Higgs standing

14 Final domain closure

The closure can be stated without technical vocabulary:

Carrier grounds identity; representatives describe or measure; scale, vacuum, extension, and metric claims classify downstream.

The formal closure is:

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Attr}_H \setminus C_H) = 0.$$

This result is the exact domain endpoint of the arc. It is stronger than a collection of non-promotion results because the residual alternative is also closed. A claim cannot avoid the endpoint merely by inventing a new Higgs-associated label; it must preserve the target, avoid every registered class, and supply a faithful non-reducing standing route. Such a submission is a burden state, and a discharged burden would revise an imported module or define a new target rather than count as an unregistered survivor inside the closed package.

15 Conclusion

The Higgs structural endpoint arc closes around a single same-scope primitive: the bridge/fixation carrier $C_H = H_{\text{br/fix}}$. The carrier does not erase standard descriptions. It licenses them. Scalar, vev, mass, potential, scale, vacuum, thermal, extension, and metric-comparison claims are classified as representatives, measurements, route statuses, scale statuses, extension dispositions, projections, target updates, counterexample burdens, or failures. They do not survive as residual primitive Higgs attributes.

Paper 6 therefore supplies the domain endpoint:

$$\text{Endpoint}_H(D_H) = (C_H, \text{Rep}_H, R_H^{\text{scale}}, R_H^{\text{vac}}, R_H^{\text{ext}}, \Pi_{\text{met}}, \text{Update}_H, \text{Fail}_H),$$

where counterexample burden is tracked by ρ_H^* and E_{burden} , not as an endpoint component or same-scope survivor. The endpoint also yields

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Attr}_H \setminus C_H) = 0.$$

That is the completed Higgs-domain standing architecture delivered by the six-paper arc.

A Theorem dependency ledger

Table 8: Paper 6 theorem dependency ledger.

Paper 6 result	Imported source	Local work
Package adequacy	Papers 1-5	Collects the closure package and verifies module coverage.
Unified same-scope normal-form theorem	$R_H^{\text{id}}, R_H^{\text{scale}}, R_H^{\text{vac}}, R_H^{\text{ext}}$	Shows every standing-bearing same-scope attribute enters the unified form through $\text{NF}_{H,\text{same}}^*$.
Layered classifier totality	Papers 2-5 classifiers plus fixed-domain update boundary	Assigns raw mode, module entry, primary disposition, and ancillary tags.
Register coverage	Papers 2-5 exhaustion registers	Unifies eliminated primitive claim forms.
No residual primitive attribute	Papers 2-5 plus projection classifier	Closes the alleged residual primitive locus.
Higgs domain endpoint	Same-scope/update normalizers, layered classifier, and no residual primitive theorem	Assembles final endpoint tuple while separating target updates from same-scope standing.
Unknown attributes theorem	Final classifier	Extends classification to future attribute submissions.
Metric projection non-primitivity	Paper 3 scale standing and projection boundary	Fixes metricity as downstream projection surface.
Counterexample burden theorem	Fixed-domain discipline and Papers 2-5	Defines faithful same-scope survivor burden.

B Unified status classifier

Table 9: Operational layered classifier table.

Question	Test	If yes	If no
Does the claim alter D_H , C_H , H_{role} , or the invariant bundle?	Boundary/update test	$\text{NF}_{H,\text{update}}^* \downarrow$, final disposition TargetUpdate	Continue.
Is the claim C_H itself?	Carrier identity test	final disposition Carrier	Continue.
Does it describe or measure C_H ?	Description or witness test	raw mode description/measurement; final disposition Representative or Measurement	Continue.

Question	Test	If yes	If no
Does it fix scale or hierarchy?	Scale normal-form test	module R_H^{scale} , final disposition ScaleStatus	Continue.
Does it concern vacuum, route, lower branch, thermal, or EWBG status?	Vacuum normal-form test	module R_H^{vac} , final disposition VacuumRouteStatus	Continue.
Does it add scalar, sector, composite, or role-splitting content without changing the boundary?	Extension normal-form test	module R_H^{ext} , final disposition ExtensionDisposition	Continue.
Does it compare through metric or gravity language?	Projection test	module Π_{met} , final disposition MetricProjection	Continue.
Does it assert a faithful non-reducing same-scope survivor?	Counterexample test	final disposition CounterexampleBurden	FailedPrimitive.

C Claim and nonclaim register

Table 10: Claim and nonclaim register.

ID	Statement	Status
P6-T1	Papers 1-5 supply the inherited Higgs endpoint package $\mathcal{P}_H^*$.	Claim
P6-T2	Every same-scope standing-bearing Higgs-domain attribute claim factors through $\mathbf{NF}_{H,\text{same}}^*$ and the unified normal form.	Claim
P6-T3	Every submitted Higgs-domain attribute claim receives layered classification by raw mode, module entry, primary disposition, and ancillary tags.	Claim
P6-T4	The cumulative exhaustion register E_H covers identity, scale, vacuum, extension, projection, update, burden, and failure routes.	Claim
P6-T5	Metric comparison is downstream projection, not primitive Higgs foundation.	Claim
P6-T6	Unknown attributes are classifier inputs, not residual primitive grounds.	Claim
P6-T7	No residual same-scope primitive Higgs attribute survives outside C_H .	Main claim
P6-T8	The Higgs domain endpoint closes as $(C_H, \mathbf{Rep}_H, R_H^{\text{scale}}, R_H^{\text{vac}}, R_H^{\text{ext}}, \Pi_{\text{met}}, \mathbf{Update}_H, \mathbf{Fail}_H)$.	Main claim
P6-T9	Target updates and counterexample burdens are submitted-claim outcomes, not same-scope standing forms.	Claim
P6-N1	The paper does not predict numerical Higgs masses, widths, self-couplings, branching ratios, or Yukawa values.	Nonclaim

ID	Statement	Status
P6-N2	The paper does not rule out BSM scalar sectors or extra scalar observations.	Nonclaim
P6-N3	The paper does not compute vacuum lifetime, instability scale, or electroweak thermal history.	Nonclaim
P6-N4	The paper does not make metricity foundational.	Nonclaim
P6-N5	The paper does not reprove the AASC kernel or Papers 1-5.	Nonclaim
P6-N6	The paper does not replace standard Higgs terminology; it classifies its standing role.	Nonclaim
P6-N7	Counterexample burden is not a standing survivor status.	Nonclaim
P6-N8	A discharged faithful counterexample revises or breaks an imported module, or declares a new target; it is not an unregistered survivor inside the closed endpoint package.	Nonclaim

D Denial-obligation register

A same-target objection must discharge at least one of the following obligations:

1. break the Paper 1 carrier package;
2. break the Paper 2 structural identity endpoint;
3. supply a primitive scale survivor not covered by Paper 3;
4. supply a primitive vacuum or lower-branch survivor not covered by Paper 4;
5. supply an extension standing survivor not covered by Paper 5;
6. show that metric projection generates same-scope Higgs standing rather than displaying carrier-indexed structure;
7. identify a submitted attribute that preserves same scope while avoiding identity-like, scale-like, vacuum-like, extension-like, metric-like, representative-like, measurement-like, target-update, counterexample-burden, and failed classifications;
8. discharge the faithful residual counterexample burden without reducing to C_H or changing D_H ;
9. show that a discharged counterexample preserves the target while breaking an imported module rather than merely entering `CounterexampleBurden`.

An objection outside these obligations changes target, refuses the proof class, or preserves the theorem chain.

E Notation index

Table 11: Notation index.

Symbol	Meaning
D_H	fixed same-scope Higgs-domain target.
C_H	Higgs bridge/fixation carrier.
H_{role}	inherited Higgs role bundle.
R_H^{id}	identity normal form.
R_H^{scale}	scale-standing normal form.
R_H^{vac}	vacuum-route normal form.
R_H^{ext}	extension normal form.
R_H^*	unified Higgs normal form.
$\text{NF}_{H,\text{same}}^*$	same-scope normalizer for standing-bearing attribute claims.
$\text{NF}_{H,\text{update}}^*$	target-update normalizer for changed-domain claims.
Π_{met}	metric/projection display layer.
$\text{Attr}_H(a)$	submitted Higgs-domain attribute claim.
$\rho_{H,\text{raw}}^*$	raw descriptive mode of an attribute claim.
$\rho_{H,\text{module}}^*$	inherited normal-form or update-branch entry.
$\rho_{H,\text{disp}}^*$	primary report-level disposition.
ρ_H^*	collective notation for the layered endpoint status classifier.
E_H	cumulative exhaustion register.
$E_{\text{update}}, E_{\text{burden}}$	update and counterexample-burden components of the cumulative exhaustion register.
$\text{Endpoint}_H(D_H)$	final Higgs-domain endpoint.

F Source support manifest

The local archive contains the previous arc manuscripts and core support papers used to substantiate the compact dependency lock. The manuscript text uses them as imported modules, not as hidden premises for new physics claims.